An Intelligent Scale Integrating Computer Vision and AI for Fruit and Vegetable Classification and Weight Measurement

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***Abstract*— This project aims to develop an embedded system that automates the weighing and recognition of fruits and vegetables for retail applications. By integrating a load cell with a camera, the system can simultaneously measure weight and classify produce using computer vision techniques. Previous research has demonstrated the effectiveness of deep learning models for fruit recognition from images, providing reliable accuracy without handcrafted features, as well as the potential of hyperspectral imaging and machine learning for robust fruit and vegetable classification under varied conditions. Additionally, applied studies have shown the feasibility of combining weight detection and visual classification to create automated checkout solutions for real-world use. Building on these advances, the proposed prototype seeks to provide a cost-effective and efficient tool to reduce human error, streamline retail operations, and enhance customer experience.**

***Keywords*— Fruit and vegetable recognition.**

I. Introduction

Automation and computer vision are rapidly transforming retail environments by reducing human error, increasing efficiency, and improving the customer experience [1]. In particular, the weighing and pricing of fruits and vegetables remain highly manual in many supermarkets, often requiring staff assistance and slowing down the checkout process. Recent works [2] have demonstrated the potential of intelligent self-checkout systems to automate this task by integrating computer vision, deep learning, and sensing technologies.

This potential is exemplified by the work [2], who achieved a recognition accuracy of 99.78% by integrating vision and weight sensors, significantly improving self-checkout reliability. Building on these advances, this project proposes the development of a low-cost prototype that combines a load cell with a camera for the automatic identification and pricing of fruits and vegetables. The feasibility of this approach is further supported by dedicated deep learning frameworks for automatic fruit recognition, with some models achieving an accuracy as high as 99.3% across various fruit types [3]. By leveraging these proven techniques, our system follows the research direction of embedded recognition solutions, aiming to provide a scalable and user-friendly approach that reduces human intervention and minimizes errors [4], [1].

Although the objective of accurate classification is common, researchers have explored various methodologies to achieve it. Some approaches, for instance, focus on a hybrid "coarse-to-fine" strategy, where weight data is first used to narrow down potential candidates before a Convolutional Neural Network (CNN) performs the final, precise visual identification [5]. Other works have tackled the practical challenge of unbalanced datasets, a common issue in retail, where some types of fruit are far more frequent than others, by developing adaptive margin-based classification models to ensure robustness [6]. These specialized techniques demonstrate that beyond simple recognition, optimizing the classification strategy for real-world retail conditions is a key area of innovation in the field.

Building upon this established potential, this study details the design, implementation, and preliminary evaluation of a functional, low-cost intelligent scale prototype. We present an architecture that decouples the real-time sensing tasks from the computationally intensive processing: a microcontroller is dedicated to acquiring precise weight data from a load cell, while a connected host computer executes an AI model for image-based fruit recognition. This work describes the integration of these hardware and software components and discusses the results of our initial tests to validate the system's core functionality.

The rest of this work is organized as follows. Section II presents the related studies and theoretical background on self-checkout systems, computer vision, and weight-based classification. Section III describes the architecture of the proposed prototype, detailing the hardware and software components. Section IV focuses on the implementation process, including the integration of the load cell and camera with the classification model. Section V analyzes the system's performance through preliminary tests and discusses the challenges encountered. Finally, Section VI concludes the work and outlines possible directions for future improvements.

II. System description

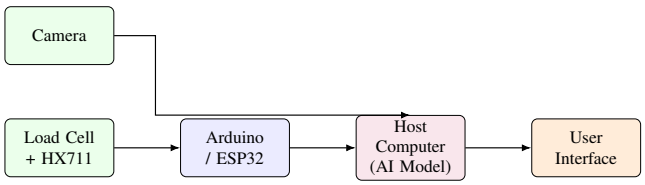

Fig. 1. General architecture of the proposed intelligent scale system.

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A. Microcontroller Platform

At the core of the system's hardware orchestration lies a microcontroller, responsible for acquiring sensor data and managing communication protocols. Boards such as Arduino Uno or ESP32 are widely adopted for embedded system prototypes due to their low power consumption, accessible development environments, and availability of digital and analog I/O. In this project, the microcontroller's primary role is to interface with the weight measurement system, retrieve the digitized weight data, and transmit it to a host computer, a common architecture in intelligent IoT-based scales [4].

B. Weight Measurement System

To ensure accurate mass measurement, the system integrates a strain gauge load cell with an HX711 signal amplifier and Analog-to-Digital Converter (ADC). The load cell functions as a transducer, converting the mechanical force exerted by a product into a minute analog electrical signal. Due to the low voltage of this signal, the HX711 module is used to perform two critical functions: it first amplifies the signal to reduce the signal-to-noise ratio, and then its high-resolution 24-bit ADC digitizes the signal for precise processing by the micro-controller. The integration of weight sensors is a fundamental component in modern self-checkout systems, often used in conjunction with visual data to improve classification accuracy [2], [5].

C. Image Acquisition System

The visual recognition capabilities of the intelligent scale are enabled by an image acquisition system which consists of a digital camera positioned to capture a clear view of the products placed on the scale. The quality of the captured image—including resolution, lighting, and focus—is a critical factor that directly influences the performance of the Artificial Intelligence model. For this prototype, the camera is interfaced with a host computer that requests an image capture each time the scale detects a new item, ensuring that the AI model receives a consistent and high-quality input for classification, a process central to automated fruit recognition frameworks [3].

D. Artificial Intelligence (AI) Model

The classification of fruits and vegetables is performed by a custom-trained Artificial Intelligence model. This software component receives an image from the acquisition system and processes it using a deep learning architecture, such as a Convolutional Neural Network (CNN), which has been proven effective for image recognition tasks [3], [5]. The model analyzes the visual features of the item, such as shape, color, and texture, to identify the specific product class. Due to the significant computational resources required for deep learning inference, the AI model is deployed on a host computer that communicates with the scale's microcontroller. Upon successful classification, the model returns a unique product identifier, a strategy similar to those used in other intelligent recognition systems for supermarkets [2], [6].

III. The Method

The development of the scale prototype began with defining the architecture and simulating the components (Arduino and HX711) in the Wokwi environment.

Upon migrating to the physical hardware, the critical step was calibration. We encountered challenges such as inverted readings (resolved by correcting the wiring) and significant fluctuations in the measurement values, as the raw sensor readings were not stable.

To achieve the necessary accuracy, we adopted the two-point calibration method. We collected the raw sensor readings for two known reference weights. With this data, we calculated the slope and offset constants and implemented them into the code to convert the raw reading into grams using the formula:

$$Weight = (slope \times Raw_Reading) + offset \tag{1}$$

Currently, the scale prototype is functional and calibrated using this two-point method, awaiting integration with the AI model.

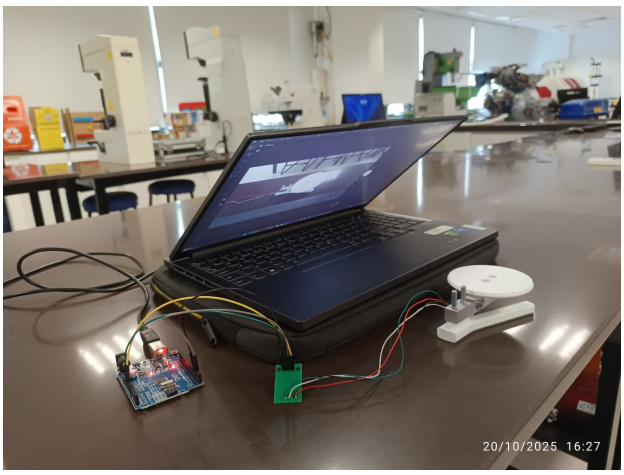

Fig. 2. Prototype

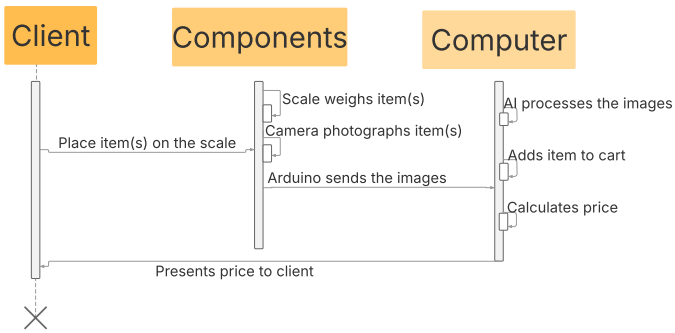

Fig. 3. Sequence diagram

The operational flow of the system, as depicted in the sequence diagram , is initiated by the Client. The process begins when the client places an item onto the scale , sending a message to the Components lifeline. This action triggers a set of internal tasks within the hardware components: the "Scale

weighs item(s)" and the "Camera photographs item(s)". Following this, the microcontroller (Arduino) is responsible for sending the captured images to the Computer. The Computer then executes its internal processing logic, which includes having the AI process the images , recognizing and registering the item (Adds item to cart), and determining the final cost (Calculates price). The cycle is concluded with the Computer presenting the final price.

IV. Results and Discussion

A. Qualitative Analysis

To validate the prototype's functionality as described in Section III, a series of qualitative evaluations were conducted. The objective of this analysis was to test the robustness of the AI model in real-world scenarios, which are known to present challenges such as variable lighting conditions, item occlusion, and human factors.

1) Analysis of Lighting Variation: The system was evaluated under three distinct lighting scenarios to simulate potential retail environments:

- **Ideal and Shadowed Conditions:** In ideal ambient light and in scenarios with partial environmental shadows, the model consistently identified items (e.g., 'apple') with high confidence, averaging between 80% and 92%.
- **Direct Light (Glare):** Direct, intense light causing reflections on the fruit's surface did not significantly impede correct detection.
- **Projected Shadow (Weakness):** A limitation was identified when the item cast a projected shadow onto the scale surface. The model occasionally attempted to classify this distinct shadow as part of the object, registering a very low-confidence detection that interfered with the primary item's classification.

2) Analysis of Position, Occlusion, and Multi-Item Detection: This analysis assessed the model's robustness to varied product presentation.

- **Position (Success):** The model demonstrated strong orientation invariance. It successfully identified the apple even when placed on its side or upside-down, indicating that the training data provided sufficient angular variety.
- **Multi-Item (Success):** The system performed exceptionally well in multi-item "clutter" scenarios. In tests featuring two distinct fruits (an apple and an orange), both objects were correctly identified and classified simultaneously. This result held true whether the items were separated or placed in direct contact. This validates the model's critical ability to handle multiple items, a key requirement for this application.
- **Severe Occlusion (Failure):** The model consistently failed to detect an item when it was partially obstructed by the user's hand. This is a clear limitation for severe occlusion cases.

3) Analysis of False Positives (Contextual Robustness): This test evaluated the system's ability to reject non-produce items, a critical function for preventing incorrect charges. Two different camera configurations were tested to assess contextual robustness.

- **Operational Context - Top-Down (90° on the zenith axis):** In the primary setup, the camera was positioned at a 90° angle directly above the scale, capturing only the weighing surface. Under these conditions, the system performed exceptionally well — it **successfully ignored all non-relevant items** such as hands, wallets, and keys. The top-down perspective not only minimizes background interference but also enhances **user privacy**, since no part of the customer's face or body is captured, making this configuration ideal for deployment in public or semi-automated retail environments.
- **Angled Context - Inclined View (45° relative to the zenith axis):** In a secondary setup, the camera was tilted to approximately 45° above the scale to simulate a more flexible or visually accessible configuration. In this case, a noticeable decrease in detection accuracy was observed. The model occasionally misclassified objects partially visible in the background or misinterpreted reflected surfaces as relevant items. This result indicates that while the system can tolerate minor variations in camera angle, maintaining a top-down orientation is crucial for both detection reliability and user privacy.

B. Quantitative Analysis

To complement the qualitative evaluation, a quantitative analysis was conducted focusing on the system's performance in terms of response time and measurement accuracy.

1) Response Time: To quantify the system's real-world usability and performance, we conducted a 10-iteration test using a 500g reference weight. Two key intervals were measured: 1) the time from placing the item on the scale until a stable weight was acquired and displayed ("Weighing Time"), and 2) the time from removing the item until the system registered zero ("Unloading Time").

The collected data yielded the following statistical results:

- **Weighing Time:** Mean (μ) = **5.70 s**; Standard Deviation (σ) = **0.85 s**
- **Unloading Time:** Mean (μ) = **10.48 s**; Standard Deviation (σ) = **1.02 s**

These results demonstrate that the current prototype achieves acceptable performance for a laboratory implementation, although additional filtering and signal stabilization could further reduce the delay in future iterations.

2) Weight Accuracy: The calibrated scale exhibited a consistent deviation when compared to a reference digital scale. Across multiple tests, the measured weight was observed to be between **1 and 4 grams above the actual value**.

This margin of error corresponds to a relative deviation of approximately 0.5–2% for items weighing around 200 g, which is considered reasonable for low-cost load cell modules such as the HX711. Further refinement of calibration constants or implementation of digital filtering techniques (e.g., moving average or exponential smoothing) could improve precision.

3) Discussion: The quantitative results confirm that the prototype operates within an acceptable range of accuracy for its intended proof-of-concept purpose. The time required

to stabilize measurements reflects the trade-off between low-cost hardware and noise filtering. Despite these limitations, the overall system responsiveness remains practical for self-checkout scenarios, particularly when integrated with the computer vision module, which processes images in parallel during the stabilization phase.

4) Illumination Condition Analysis: Following the qualitative observations discussed earlier, a quantitative evaluation was conducted to measure how illumination intensity affects the model's detection accuracy. Light levels were recorded using a mobile lux meter app, and the same fruit samples were tested under each condition. Table I summarizes the results.

TABLE I. Detection accuracy under different illumination conditions.

Illumination (lux)	Environment	Accuracy (%)
100	Low	66.7
300	Moderate	83.3
700	Bright	96.7
1200	Intense	93.3

The results indicate that detection accuracy improves significantly as illumination increases up to around 700 lux. Under poor lighting (below 300 lux), the model fails to clearly identify fruit contours. However, at excessive brightness levels (above 1000 lux), performance slightly decreases due to reflections and shadows. Therefore, the optimal lighting range for stable operation lies between 500 and 800 lux.

5) AI Model Accuracy Evaluation: To assess the detection performance of the computer vision component, an F1 versus threshold curve was generated for the YOLO-based fruit detection model, as shown in Figure 4.

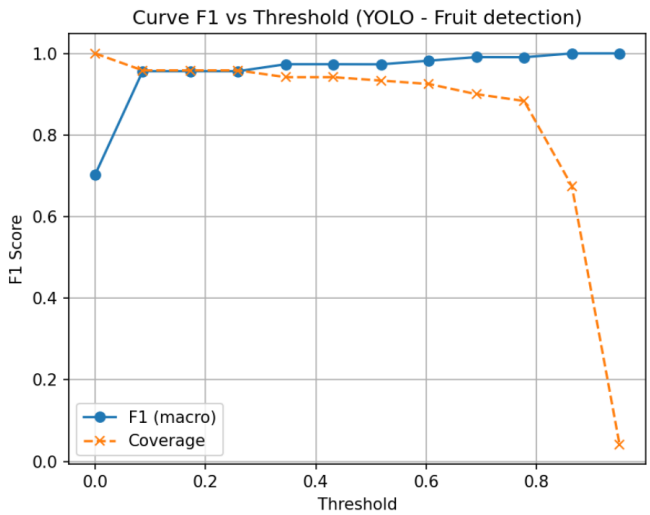

Fig. 4. F1 vs Threshold curve for the YOLO-based fruit detection model.

The curve illustrates the model's F1 score and coverage across confidence thresholds ranging from 0.4 to 0.95. Results show that the model maintains a high and stable performance (F1 $\approx$ 1.0) between thresholds 0.4 and 0.8, indicating strong robustness in identifying fruit objects under ideal operational conditions. However, beyond a threshold of 0.8, the coverage drops sharply, revealing that the system becomes overly selective, leading to missed detections.

These findings suggest that a confidence threshold around 0.8 represents the best trade-off between precision and recall for this prototype, balancing detection reliability and generalization.

V. Conclusions

This work presented the design and implementation of an intelligent scale that integrates weight measurement and computer vision for the automatic recognition of fruits and vegetables. The prototype successfully combines a load cell, HX711 amplifier, and a camera connected to a host computer running an AI-based classification model.

The qualitative tests demonstrated that the system performs robustly under different lighting conditions and orientations, correctly identifying multiple items and rejecting non-produce objects in its operational context. The quantitative analysis confirmed that the scale operates with a stabilization time between 7 and 8.5 seconds for weighing and 4 to 15 seconds for unloading, and a deviation of 1 to 4 grams from the actual weight.

These results validate the feasibility of a low-cost intelligent weighing system capable of integrating into self-checkout environments. Future improvements will focus on refining calibration accuracy, reducing response time, and integrating the AI model directly into an embedded platform such as the Raspberry Pi, enhancing autonomy and scalability for commercial applications.

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